Log-polyhomogeneous pseudodifferential operators on filtered manifolds and their spectral asymptotics

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May 27, 2026

Introduction

In this talk I will discuss the tangent groupoid theory of pseudodifferential operators whose symbols have asymptotic expansion like

$$\sigma(x,\xi) \sim \sum_{\ell=0}^{k}\sum_{j=0}^{\infty} a_{j,\ell}(x,\xi)\log(|\xi|)^{\ell}$$

as $\xi \to \infty$, where $a_{j,k}(x,\xi)$ is homogeneous of order $m-j$ in ξ.

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as $\xi \to \infty$, where $a_{j,k}(x,\xi)$ is homogeneous of order $m-j$ in ξ.
Log-polyhomogeneous pseudodifferential operators were introduced by Lesch in 1999.

The main theorem

Theorem

Let (M, H) be a compact filtered manifold, with homogeneous dimension Q. Let A be a k-log-polyhomogeneous pseudodifferential operator of order $-Q$. Let $\{\lambda(n, A)\}_{n=0}^{\infty}$ be the sequence of eigenvalues of A, arranged in order of decreasing absolute value. We have

$$\sum_{n=0}^{N} \lambda(n, A) = \frac{1}{Q(k+1)!} \mathrm{Res}_k(A) \log(N)^{k+1} + o(\log(N)^{k+1}), \quad N \to \infty.$$

I will explain the notation and terminology later.

Section 1: What is a log-polyhomogeneous function?

Some notation

Let $V = \mathbb{R}^N$ be a graded vector space, with anisotropic dilations

$$\delta_t = \bigoplus_{k=1}^{N} t^{w_k}, \quad t > 0.$$

where $w_1, \ldots, w_N > 0$.

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$$\delta_t = \bigoplus_{k=1}^{N} t^{w_k}, \quad t > 0.$$

where $w_1, \ldots, w_N > 0$.
δ_t acts on distributions $K \in \mathcal{S}'(V)$ by pushfoward.

$$((\delta_t)^* K, f) = (K, f \circ \delta_t^{-1}), \quad t > 0,\ f \in \mathcal{S}(V).$$

Homogeneous and almost-homogeneous functions

We say that $K \in \mathcal{S}'(V)$ is *homogeneous* of degree $m \in \mathbb{C}$ if

$$(\delta_t)^* K = t^m K, \quad t > 0$$

and *almost homogeneous* of degree $m \in \mathbb{C}$ if

$$(\delta_t)^* K - t^m K \in \mathcal{S}(V), \quad t > 0.$$

Infinitesimal forms of homogeneity

Let $Zf := \frac{d}{dt}|_{t=1} f \circ \delta_t$ be the generator of the dilations δ_t, i.e.

$$Z := \sum_{j=1}^{N} w_j x_j \frac{\partial}{\partial x_j}$$

Theorem (Euler, 18th century)

A distribution $K \in \mathcal{S}'(V)$ is homogeneous of degree m if and only if

$$(Z - m)K = 0.$$

Infinitesimal form of almost-homogeneity

Similarly...

Theorem

A distribution $K \in \mathcal{S}'(V)$ is almost-homogeneous of degree m if and only if

$$(Z - m)K \in \mathcal{S}(V).$$

Taylor's lemma

Taylor proved that an almost-homogeneous function is almost a homogeneous function.

Lemma (Taylor(?), 1984)

Let $K \in C^\infty(V) \cap \mathcal{S}'(V)$. If K is almost homogeneous of degree m, then there exists a homogeneous function $H \in C^\infty(V \setminus \{0\})$ such that for any $\psi \in C_c^\infty(V)$ with $\psi = 1$ near zero, we have

$$K - (1-\psi)H \in \mathcal{S}(V).$$

Obviously we must have

$$H(\xi) = \lim_{t\to\infty} t^{-m} K(\delta_t \xi), \quad x \in V \setminus \{0\}.$$

The theorem more-or-less amounts to the identity

$$t^{-m}K(\delta_t\xi) - K(\xi) = \int_1^t s^{-m-1}((Z-m)K)(\delta_s\xi)\, ds, \quad t > 0, \xi \in \mathbb{R}^N.$$

Polyhomogeneous functions

A function $K \in C^\infty(V)$ is *polyhomogeneous* if there exist homogeneous functions $\{K_\ell\}_{\ell=0}^\infty$ of degrees $\{m-\ell\}_{\ell=0}^\infty$ such that

$$K(\xi) \sim \sum_{\ell=0}^\infty K_\ell(\xi), \quad |\xi| \to \infty$$

in the usual sense.

Theorem (Debord-Skandalis, 2014)

Let $\widetilde{\delta}_t(\xi, h) := (\delta_t \xi, th)$. A smooth function K on V is polyhomogeneous of order m if and only if there exists an almost homogeneous function $\widetilde{K}$ on $V \times \mathbb{R}$ such that

$$K(\xi) = \widetilde{K}(\xi, 1), \quad \xi \in \mathbb{R}^N.$$

Moreover, $\widetilde{K}(\xi, 0)$ is an almost-homogeneous function whose homogeneous part is the principal term K_0.

Log-homogeneous functions

Let $|\cdot|$ be a δ-homogeneous norm on V, i.e. $|\delta_t\xi| = t|\xi|$.
Functions of the form $L(\xi) = \log(|\xi|)$ are special because $L\circ\delta_t = \log(t) + L$, or in other words

$$ZL = 1.$$

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We will say that a function $K \in C^\infty(V \setminus \{0\})$ is log-homogeneous of order (m, k) if there exist $H_0, \ldots, H_k$,homogeneous functions of order m, such that

$$K(\xi) = \sum_{\ell=0}^{k} H_\ell(\xi) \log(|\xi|)^\ell.$$

Log-homogeneous functions

Lemma

A function $K \in C^\infty(V \setminus \{0\})$ *is log-homogeneous of order* (m, k) *if and only*

$$(Z - m)^{k+1} K = 0.$$

Proof: Induction on k.

Almost-log-homogeneous functions

By analogy with almost-homogeneous functions, say that $K \in C^{\infty}(V)$ is almost log-homogeneous of order (m, k) if

$$(Z - m)^{k+1} K \in \mathcal{S}(V)$$

Lemma (Logarithmic Taylor lemma)

Let $K \in C^{\infty}(V) \cap \mathcal{S}'(V)$. If K is almost homogeneous of degree m, then there exists a log-homogeneous function $H \in C^{\infty}(V \setminus \{0\})$ such that for any $\psi \in C_c^{\infty}(V)$ with $\psi = 1$ near zero, we have

$$K - (1 - \psi) H \in \mathcal{S}(V).$$

log-polyhomogeneous functions

Say that a function $K \in C^\infty(V)$ is log-polyhomogeneous of order (m,k) if there exist homogeneous functions $\{K_{\ell,j}\}_{j=0}^\infty$ such that

$$K(\xi) \sim \sum_{\ell=0}^{k} \sum_{j=0}^{\infty} \log(|\xi|)^j K_{\ell,j}(\xi), \quad |\xi| \to \infty.$$

Theorem

Let $\widetilde{\delta}_t(\xi,h) := (\delta_t\xi, th)$. A smooth function K on V is log-polyhomogeneous of order (m,k) if and only if there exists an almost homogeneous function $\widetilde{K}$ on $V \times \mathbb{R}$ such that

$$K(\xi) = \widetilde{K}(\xi,1), \quad \xi \in \mathbb{R}^N.$$

Moreover, $\widetilde{K}(\xi,0)$ is an almost log-homogeneous function whose log-homogeneous part is the principal term $\xi \to \log(|\xi|)^k K_{k,0}(\xi)$.

Section 2: What is an operator with log-polyhomogeneous symbol?

The H-tangent groupoid

Let M be a manifold, and let

$$0 = H^0 < H^1 < \cdots < H^N = TM$$

be a Lie filtration of the tangent bundle. We have an osculating bundle $T_H M$ of graded nilpotent Lie groups and an H-tangent groupoid

$$\mathbb{T}_H M := (T_H M \times \{0\}) \sqcup (M \times M \times \mathbb{R}^\times).$$

which is equipped with the correct topology.

Reminder on notation

- (M, H) is a filtered manifold.
- $Q = \sum_{k=1}^{\infty} k(\mathrm{rank}(H^k) - \mathrm{rank}(H^{k-1}))$ is its homogeneous dimension.
- $T_H M$ is the osculating groupoid, with range map

$$r : T_H M \to M$$

- $\mathbb{T}_H M$ is the H-tangent groupoid, with range map

$$r : \mathbb{T}_H M \to \mathbb{R} \times M.$$

- $\mathcal{E}'_r(G)$ is the space of r-fibred distributions

$$C^\infty(G) \to C^\infty(G^{(0)})$$

- $\Omega_r := \Omega\mathrm{ker}(dr)$ is the bundle of densities transversal to the r-fibres.
- $C_p^\infty(G, \Omega_r)$ is the space of r and s properly supported r-transversal densities.

The H-tangent groupoid

The zoom action on $\mathbb{T}_H M$ is

$$\alpha_t(x, y, h) = (x, y, t^{-1}h), \quad \alpha_t((x, z), 0) = ((x, \delta_t z), 0).$$

The generator of the zoom action

Let Z be the vector field on $\mathbb{T}_H M$ which generates the zoom action, i.e.

$$Zf := \frac{d}{dt}|_{t=1} f \circ \alpha_t.$$

The Lie derivative by Z acts on r-fibred distributions, i.e.

$$\mathcal{L}_Z \mathbb{P} := \frac{d}{dt}_{t=1} (\alpha_t)_* \mathbb{P}.$$

The van Erp-Yuncken calculus

Definition (van Erp-Yuncken (2019))

Let

$$\Psi_H^m(M) = \{T \in \mathcal{E}_r'(\mathbb{T}_H M) \ : \ (\mathcal{L}_Z - m)\mathbb{P} \in C_p^\infty(\mathbb{T}_H M, \Omega_r)\}.$$

A pseudodifferential operator P of order m is defined by its Schwartz kernel being of the form $\mathbb{P}_{h=1}$ for some pseudodifferential family $\mathbb{P}$.

Moreover, if $\mathbb{P}_1 = P$, then $\mathbb{P}_0$ is an approximately homogeneous distribution on $T_H M$ which represents the principal (co)symbol of P.

The log-polyhomogeneous van Erp-Yuncken calculus

The natural thing to do is to define k-log-polyhomogeneous operators in the same way.

Definition

Let

$$\Psi_H^{m,k}(M) = \{T \in \mathcal{E}_r'(\mathbb{T}_H M) \; : \; (\mathcal{L}_Z - m)^{k+1}\mathbb{P} \in C_p^\infty(\mathbb{T}_H M, \Omega_r)\}.$$

A pseudodifferential operator P of order (m,k) is defined by its Schwartz kernel being of the form $\mathbb{P}_{h=1}$ for some pseudodifferential family $\mathbb{P}$.

Denote the set of pseudodifferential operators with k-log-polyhomogeneous order (m,k) as $\Psi_H^{m,k}(M)$.

The log-polyhomogeneous van Erp-Yuncken calculus

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A pseudodifferential operator P of order (m,k) is defined by its Schwartz kernel being of the form $\mathbb{P}_{h=1}$ for some pseudodifferential family $\mathbb{P}$.

Denote the set of pseudodifferential operators with k-log-polyhomogeneous order (m,k) as $\Psi_H^{m,k}(M)$.
We get nice clean statements if we define

$$\mathbb{\Psi}_H^{m,-1}(M) = C_p^\infty(\mathbb{T}_H M, \Omega_r), \quad \Psi_H^{m,-1}(M) = C^\infty(M \times M, \Omega M).$$

Yes, it is the right definition

Theorem (M. 2026)

If $H = \mathrm{triv.}$ *is the trivial filtration, then* $\Psi_H^{m,k}(M)$ *is the operators which locally have symbol function that is log-polyhomogeneous of order* (m,k).

Some obvious facts

It is pretty straightforward to check that $\bigcup_{m,k} \Psi_H^{m,k}(M)$ is bi-graded by m and k, i.e.

$$\Psi_H^{m_0,k_0}(M) \cdot \Psi_H^{m_1,k_1}(M) \subseteq \Psi_H^{m_0+m_1,k_0+k_1}(M).$$

We also have closure under the adjoint operation, etc. etc.

Uniqueness of pseudodifferential families

Lemma

Let $\mathbb{P} \in \Psi_H^{m,k}(M)$. *If* $\mathbb{P}_1 = 0$, *then* $\mathbb{P} \in \Psi_H^{m,k-1}(M)$.

A less-than-obvious fact

The symbol mapping goes like

$$0 \to \Psi_H^{m-1,k}(M) \to \Psi_H^{m,k}(M) \to \Sigma_H^{m,k}(M) \to 0.$$

where the symbol class $\Sigma_H^{m,k}(M)$ has a non-obvious definition.

Definition

Let

$$\Sigma_H^{m,k}(M) := \{T \in \mathcal{E}_r'(T_H M) \,:\, (\mathcal{L}_Z - m)^{k+1} T \in C_p^\infty(T_H M, \Omega_r)\}, \quad k \geq 0$$

and

$$\Sigma_H^{m,k}(M) := \Sigma_H^{m,k}(M)/\Sigma_H^{m,k-1}(M), \quad k \geq 1.$$

Section 3: The residue and the trace

Back to functions

Again let V be a vector space with a group of dilations $\{\delta_t\}_{t>0}$. Let $S^m_{\mathrm{phg}}(V)$ be the set of polyhomogeneous smooth functions of order m, i.e.

$$S^m_{\mathrm{phg}}(V) = \{f \in C^\infty(V) \;:\; f \sim \sum_{j=0}^\infty f_j,\; f_j(\delta_t \xi) = t^m f_j(\xi)\}.$$

There are two important functionals on $S^m_{\mathrm{phg}}(V)$,, depending on the value of m. They are the *canonical integral*,

$$\mathrm{Int} : S^m_{\mathrm{phg}}(V) \to \mathbb{C}, \quad m \in \mathbb{C} \setminus \{-Q, -Q+1, -Q+2, \ldots\}$$

and the *residue*

$$\mathrm{Res} : S^m_{\mathrm{phg}}(V) \to \mathbb{C}, \quad m \in \mathbb{Z}.$$

Debord-Skandalis description of the canonical integral and the residue

Let $K \in S^m_{\mathrm{phg}}(V)$. By Debord-Skandalis' characterisation, there exists an almost-homogeneous function $\widetilde{K}$ on $V \times \mathbb{R}$ such that

$$K(\xi) = \widetilde{K}(\xi, 1).$$

Let

$$a(\xi, h) := \frac{d}{dt}|_{t=1}(t^{-m}\widetilde{K} \circ \widetilde{\delta}_t) = (\widetilde{Z} - m)\widetilde{K}(\xi, h).$$

and $r(h) := \int_V a(\xi, h)\, d\xi$. We have

$$\mathrm{Res}(K) = r(0)$$

and for sufficiently large n,

$$\mathrm{Int}(K) = \sum_{j=0}^{n} \frac{1}{j!} \frac{r^{(j)}(0)}{j - m - Q}$$

$$+ \frac{1}{(n-1)!} \int_0^1 r^{(n)}(u)\, u^{n-m-Q-1} \mathrm{B}(1 - u; n, m + Q - n + 1)\, du$$

Couchet-Yuncken's residue

Let

$$\tau : C_p^\infty(\mathbb{T}_H M, \Omega_r) \to C^\infty(M, \Omega M)$$

be the evaluation at $h = 0$.

Definition

For $A \in \Psi_H^{-Q}(M)$, let

$$\mathrm{res}(A) := \tau((\mathcal{L}_Z + Q)\mathbb{A}), \quad \mathbb{A}_1 = A.$$

Define $\mathrm{Res}(A) := \int_M \mathrm{res}(A)$.

Definition

For $A \in \Psi_H^{-Q,k}(M)$, let

$$\mathrm{res}_k(A) := \tau((\mathcal{L}_Z + Q)^{k+1}\mathbb{A}), \quad \mathbb{A}_1 = A.$$

Define $\mathrm{Res}_k(A) = \int_M \mathrm{res}_k(A)$.

Section 4: Spectral theory and the trace theorem

Operator ideals

Let $\mathcal{H}$ be a Hilbert space, and $\mathcal{K}(\mathcal{H})$ is the ideal of compact operators on $\mathcal{H}$. For $A \in \mathcal{K}(\mathcal{H})$, let $\lambda(A) = \{\lambda(n, A)\}_{n=0}^{\infty}$ be a sequence of eigenvalues of A.

1. We arrange the sequence such that $|\lambda(n, A)|$ is non-increasing. This is ambiguous, but ultimately unimportant.
2. If A has finitely many eigenvalues (including none at all, this can happen in the non-normal case!) then $\lambda(A)$ should be filled out with zeros.

Weyl's inequality

$\lambda(n,|A|)$ is the $(n+1)$th singular value of A. Note that $|\lambda(n,A)|$ and $\lambda(n,|A|)$ are very different in general. The best we can say is that

$$\prod_{n=0}^{N} |\lambda(n,A)| \leq \prod_{n=0}^{N} \lambda(n,|A|), \quad N \geq 0.$$

(This is Weyl's inequality).

Logarithmically weak-trace class operators

Definition

Let $p, k \geq 0$. Define

$$\mathcal{E}_{p,k}(\mathcal{H}) := \{A \in \mathcal{K}(\mathcal{H}) \;:\; \lambda(n, |A|) = O(\frac{\log(n)^k}{n^{1/p}})\}.$$

Elementary eigenvalue estimates ensure that

$$\mathcal{E}_{p,k} \cdot \mathcal{E}_{q,\ell} \subseteq \mathcal{E}_{r,k+\ell}, \quad r^{-1} = p^{-1} + q^{-1}.$$

Log-polyhomogeneous operators are weak-trace-class

Theorem (M. (2026))

Let (M, H) be a closed filtered manifold. For $m > 0$, we have

$$\Psi_H^{-m,k}(M) \subset \mathcal{E}_{\frac{Q}{m},k}(L_2(M)).$$

How is it proved?

For any compact operator A, we have

$$\lambda(n, |A|) := \inf\{\|A - R\|_\infty \ : \ \mathrm{rank}(R) \leq n\}.$$

To estimate $\lambda(n, |A|)$, we need:

1. A candidate sequence of finite-rank approximations to A, $\{R_n\}_{n=0}^{\infty}$,
2. A good estimate for the norms $\|A - R_n\|_\infty$.

Debord-Skandalis representation

Let $\mathbb{P} \in \Psi_H^{m,k}(M)$. If $m < 0$, we have

$$\mathbb{P} = \frac{1}{k!} \int_0^1 t^{-m-1} \log(t)^k (\alpha_t)_*((\mathcal{L}_Z - m)^{k+1}\mathbb{P})\, dt.$$

This gives a candidate family of smoothing operators which approximate $\mathbb{P}_1$.

Eigenvalue estimates for weak-trace-class operators

Weyl's inequality implies that if $A \in \Psi_H^{m,k}(M)$ with $m < 0$, then

$$\lambda(n, A) = O(n^{\frac{Q}{m}} \log(n)^k), \quad n \to \infty.$$

Probably we can do better than this, and get

$$\lambda(n, A) \sim c(A) n^{\frac{Q}{m}} \log(n)^k, \quad n \to \infty.$$

for some $c(A)$, but this is out of reach at the moment.

Asymptotics for sums of eigenvalues

The best I can do at the moment concerns $m = -Q$ and spectral averages:

Theorem (M. (2026))

Let $A \in \Psi_H^{-Q,k}(M)$. We have

$$\sum_{n=0}^{N} \lambda(n, A) = \frac{1}{(k+1)!Q} \mathrm{Res}_k(A) \log(N)^{k+1} + o(\log(N)^{k+1}).$$

Dixmier trace reformulation

It is helpful to use the Dixmier traces

$$\mathrm{tr}_{\omega,k}(A) = \omega\left(\left\{\frac{1}{\log(N+2)^k}\sum_{n=0}^{N}\lambda(n,A)\right\}_{k=0}^{\infty}\right)$$

The theorem says that

$$\mathrm{tr}_{\omega,k}(A) = \frac{1}{(k+1)!Q}\mathrm{Res}_k(A).$$

Thank you for listening!